

```
import pandas as pd
import numpy as np
```

```
movies = pd.read_csv('/content/imdb-top-1000.csv')
```

```
movies.head()
```

	Series_Title	Released_Year	Runtime	Genre	IMDB_Rating	Director	Star1	No_of_Votes	Gross	Metascore
0	The Shawshank Redemption	1994	142	Drama	9.3	Frank Darabont	Tim Robbins	2343110	28341469.0	80.0
1	The Godfather	1972	175	Crime	9.2	Francis Ford Coppola	Marlon Brando	1620367	134966411.0	100.0
2	The Dark Knight	2008	152	Action	9.0	Christopher Nolan	Christian Bale	2303232	534858444.0	84.0
3	The Godfather Part II	1974	197	Crime	9.2	Francis Ford Coppola	Al Pacino	1600000	57600000.0	92.0

```
genres = movies.groupby('Genre')
```

```
# Applying builtin aggregation fuctions on groupby objects
genres.std()
```

	Runtime	IMDB_Rating	No_of_Votes	Gross	Metascore
Genre					
Action	28.500706	0.304258	432946.814748	2.256724e+08	12.421252
Adventure	33.317320	0.229781	301188.347642	1.697543e+08	12.345393
Animation	14.530471	0.253221	262173.231571	2.091840e+08	8.813646
Biography	25.514466	0.267140	271284.191372	1.363251e+08	11.028187
Comedy	22.946213	0.228771	188653.570564	1.946513e+08	11.829160
Crime	27.689231	0.335477	373999.730656	1.571191e+08	13.099102
Drama	27.740490	0.267229	305554.162841	2.201164e+08	12.744687
Family	10.606602	0.000000	137008.302816	3.048412e+08	16.970563
Fantasy	12.727922	0.141421	22179.111299	7.606861e+07	NaN
Film-Noir	4.000000	0.152753	54649.083277	7.048472e+07	1.527525
Horror	13.604812	0.311302	234883.508691	9.965017e+07	15.362291
Mystery	14.475423	0.310791	404621.915297	1.567524e+08	18.604435
Thriller	NaN	NaN	NaN	NaN	NaN
Western	17.153717	0.420317	263489.554280	1.230626e+07	9.032349

```
# find the top 3 genres by total earning
movies.groupby('Genre').sum()['Gross'].sort_values(ascending=False).head(3)
```

```
Genre
Drama      3.540997e+10
Action     3.263226e+10
Comedy     1.566387e+10
Name: Gross, dtype: float64
```

```
movies.groupby('Genre')['Gross'].sum().sort_values(ascending=False).head(3)
```

```
Genre
Drama      3.540997e+10
Action     3.263226e+10
Comedy     1.566387e+10
Name: Gross, dtype: float64
```

```
# find the genre with highest avg IMDB rating
movies.groupby('Genre')['IMDB_Rating'].mean().sort_values(ascending=False).head(1)
```

```
Genre
Western      8.35
```

Name: IMDB_Rating, dtype: float64

```
# find director with most popularity
movies.groupby('Director')['No_of_Votes'].sum().sort_values(ascending=False).head(1)
```

Director
Christopher Nolan 11578345
Name: No_of_Votes, dtype: int64

```
# find the highest rated movie of each genre
# movies.groupby('Genre')['IMDB_Rating'].max()
```

Genre
Action 9.0
Adventure 8.6
Animation 8.6
Biography 8.9
Comedy 8.6
Crime 9.2
Drama 9.3
Family 7.8
Fantasy 8.1
Film-Noir 8.1
Horror 8.5
Mystery 8.4
Thriller 7.8
Western 8.8
Name: IMDB_Rating, dtype: float64

```
# find number of movies done by each actor
# movies['Star1'].value_counts()
```

```
movies.groupby('Star1')['Series_Title'].count().sort_values(ascending=False)
```

Star1
Tom Hanks 12
Robert De Niro 11
Clint Eastwood 10
Al Pacino 10
Leonardo DiCaprio 9
..
Glen Hansard 1
Giuseppe Battiston 1
Giulietta Masina 1
Gerardo Taracena 1
Ömer Faruk Sorak 1
Name: Series_Title, Length: 660, dtype: int64

```
# GroupBy Attributes and Methods
# find total number of groups -> len
# find items in each group -> size
# first()/last() -> nth item
# get_group -> vs filtering
# groups
# describe
# sample
# nunique
```

```
len(movies.groupby('Genre'))
```

14

```
movies['Genre'].nunique()
```

14

```
movies.groupby('Genre').size()
```

Genre
Action 172
Adventure 72
Animation 82
Biography 88
Comedy 155
Crime 107
Drama 289

```
Family      2
Fantasy     2
Film-Noir   3
Horror      11
Mystery     12
Thriller    1
Western     4
dtype: int64
```

```
genres = movies.groupby('Genre')
# genres.first()
# genres.last()
genres.nth(6)
```

↔

	Series_Title	Released_Year	Runtime	IMDB_Rating	Director	Star1	No_of_Votes	Gross	Metascore
Genre									
Action	Star Wars: Episode V - The Empire Strikes Back	1980	124	8.7	Irvin Kershner	Mark Hamill	1159315	290475067.0	82.0
Adventure	North by Northwest	1959	136	8.3	Alfred Hitchcock	Cary Grant	299198	13275000.0	98.0
Animation	WALL·E	2008	98	8.4	Andrew Stanton	Ben Burtt	999790	223808164.0	95.0
Biography	Braveheart	1995	178	8.3	Mel Gibson	Mel Gibson	959181	75600000.0	68.0
Comedy	The Great Dictator	1940	125	8.4	Charles Chaplin	Charles Chaplin	203150	288475.0	NaN
Crime	Se7en	1995	127	8.6	David Fincher	Morgan Freeman	1445096	100125643.0	65.0
Drama	It's a Wonderful Life	1946	130	8.6	Frank Capra	James Stewart	405801	82385199.0	89.0

```
movies['Genre'].value_counts()
```

↔

```
Drama      289
Action     172
Comedy     155
Crime      107
Biography   88
Animation   82
Adventure   72
Mystery     12
Horror      11
Western     4
Film-Noir   3
Fantasy     2
Family       2
Thriller    1
Name: Genre, dtype: int64
```

```
genres.get_group('Fantasy')
```

```
movies[movies['Genre'] == 'Fantasy']
```

↔

	Series_Title	Released_Year	Runtime	Genre	IMDB_Rating	Director	Star1	No_of_Votes	Gross	Metascore
321	Das Cabinet des Dr. Caligari	1920	76	Fantasy	8.1	Robert Wiene	Werner Krauss	57428	337574718.0	NaN
568	Nosferatu	1922	94	Fantasy	7.9	F.W. Murnau	Max Schreck	88794	445151978.0	NaN

```
genres.groups
```

↔

```
{ 'Action': [2, 5, 8, 10, 13, 14, 16, 29, 30, 31, 39, 42, 44, 55, 57, 59, 60, 63, 68, 72, 106, 109, 129, 130, 134, 140, 142, 144, 152, 155, 160, 161, 166, 168, 171, 172, 177, 181, 194, 201, 202, 216, 217, 223, 224, 236, 241, 262, 275, 294, 308, 320, 325, 326, 331, 337, 339, 340, 343, 345, 348, 351, 353, 356, 357, 362, 368, 369, 375, 376, 390, 410, 431, 436, 473, 477, 479, 482, 488, 493, 496, 502, 507, 511, 532, 535, 540, 543, 564, 569, 570, 573, 577, 582, 583, 602, 605, 608, 615, 623, ...], 'Adventure': [21, 47, 93, 110, 114, 116, 118, 137, 178, 179, 191, 193, 209, 226, 231, 247, 267, 273, 281, 300, 301, 304, 306, 323, 329, 361, 366, 377, 402, 406, 415, 426, 458, 470, 497, 498, 506, 513, 514, 537, 549, 552, 553, 566, 576, 604, 609, 618, 638, 647, 675, 681, 686, 692, 711, 713, 739, 755, 781, 797,
```

```
798, 851, 873, 884, 912, 919, 947, 957, 964, 966, 984, 991], 'Animation': [23, 43, 46, 56, 58, 61, 66, 70, 101, 135, 146, 151, 158,
170, 197, 205, 211, 213, 219, 229, 230, 242, 245, 246, 270, 330, 332, 358, 367, 378, 386, 389, 394, 395, 399, 401, 405, 409, 469, 499,
510, 516, 518, 522, 578, 586, 592, 595, 596, 599, 633, 640, 643, 651, 665, 672, 694, 728, 740, 741, 744, 756, 758, 761, 771, 783, 796,
799, 822, 828, 843, 875, 891, 892, 902, 906, 920, 956, 971, 976, 986, 992], 'Biography': [7, 15, 18, 35, 38, 54, 102, 107, 131, 139,
147, 157, 159, 173, 176, 212, 215, 218, 228, 235, 243, 263, 276, 282, 290, 298, 317, 328, 338, 342, 346, 359, 360, 365, 372, 373, 385,
411, 416, 418, 424, 429, 484, 525, 536, 542, 545, 575, 579, 587, 600, 606, 614, 622, 632, 635, 644, 649, 650, 657, 671, 673, 684, 729,
748, 753, 757, 759, 766, 770, 779, 809, 810, 815, 820, 831, 849, 858, 877, 882, 897, 910, 915, 923, 940, 949, 952, 987], 'Comedy': [19,
26, 51, 52, 64, 78, 83, 95, 96, 112, 117, 120, 127, 128, 132, 153, 169, 183, 192, 204, 207, 208, 214, 221, 233, 238, 240, 250, 251,
252, 256, 261, 266, 277, 284, 311, 313, 316, 318, 322, 327, 374, 379, 381, 392, 396, 403, 413, 414, 417, 427, 435, 445, 446, 449, 455,
459, 460, 463, 464, 466, 471, 472, 475, 481, 490, 494, 500, 503, 509, 526, 528, 530, 531, 533, 538, 539, 541, 547, 557, 558, 562, 563,
565, 574, 591, 593, 594, 598, 613, 626, 630, 660, 662, 667, 679, 680, 683, 687, 701, ...], 'Crime': [1, 3, 4, 6, 22, 25, 27, 28, 33,
37, 41, 71, 77, 79, 86, 87, 103, 108, 111, 113, 123, 125, 133, 136, 162, 163, 164, 165, 180, 186, 187, 189, 198, 222, 232, 239, 255,
257, 287, 288, 299, 305, 335, 363, 364, 380, 384, 397, 437, 438, 441, 442, 444, 450, 451, 465, 474, 480, 485, 487, 505, 512, 519, 520,
523, 527, 546, 556, 560, 584, 597, 603, 607, 611, 621, 639, 653, 664, 669, 676, 695, 708, 723, 762, 763, 767, 775, 791, 795, 802, 811,
823, 827, 833, 885, 895, 921, 922, 926, 938, ...], 'Drama': [0, 9, 11, 17, 20, 24, 32, 34, 36, 40, 45, 50, 53, 62, 65, 67, 73, 74, 76,
80, 82, 84, 85, 88, 89, 90, 91, 92, 94, 97, 98, 99, 100, 104, 105, 121, 122, 124, 126, 138, 141, 143, 148, 149, 150, 154, 156, 167,
174, 175, 182, 184, 185, 188, 190, 195, 196, 199, 200, 203, 206, 210, 225, 227, 234, 237, 244, 248, 249, 253, 254, 258, 259, 260, 264,
265, 268, 269, 272, 274, 278, 279, 280, 283, 285, 286, 289, 291, 292, 293, 295, 296, 297, 302, 303, 307, 310, 312, 314, 315, ...],
'Family': [688, 698], 'Fantasy': [321, 568], 'Film-Noir': [309, 456, 712], 'Horror': [49, 75, 271, 419, 544, 707, 724, 844, 876, 932,
948], 'Mystery': [69, 81, 119, 145, 220, 393, 420, 714, 829, 899, 959, 961], 'Thriller': [700], 'Western': [12, 48, 115, 691]}
```

```
genres.describe()
```



	Runtime								IMDB_Rating		...	Gross		Metascore	
	count	mean	std	min	25%	50%	75%	max	count	mean	...	75%	max	count	mean
		Genre													
Action	172.0	129.046512	28.500706	45.0	110.75	127.5	143.25	321.0	172.0	7.949419	...	2.674437e+08	936662225.0	143.0	73.41958
Adventure	72.0	134.111111	33.317320	88.0	109.00	127.0	149.00	228.0	72.0	7.937500	...	1.998070e+08	874211619.0	64.0	78.43750
Animation	82.0	99.585366	14.530471	71.0	90.00	99.5	106.75	137.0	82.0	7.930488	...	2.520612e+08	873839108.0	75.0	81.09333
Biography	88.0	136.022727	25.514466	93.0	120.00	129.0	146.25	209.0	88.0	7.938636	...	9.829924e+07	753585104.0	79.0	76.24050
Comedy	155.0	112.129032	22.946213	68.0	96.00	106.0	124.50	188.0	155.0	7.901290	...	8.107809e+07	886752933.0	125.0	78.72000
Crime	107.0	126.392523	27.689231	80.0	106.50	122.0	141.50	229.0	107.0	8.016822	...	7.102163e+07	790482117.0	87.0	77.08046
Drama	289.0	124.737024	27.740490	64.0	105.00	121.0	137.00	242.0	289.0	7.957439	...	1.164461e+08	924558264.0	241.0	79.70124
Family	2.0	107.500000	10.606602	100.0	103.75	107.5	111.25	115.0	2.0	7.800000	...	3.273329e+08	435110554.0	2.0	79.00000
Fantasy	2.0	85.000000	12.727922	76.0	80.50	85.0	89.50	94.0	2.0	8.000000	...	4.182577e+08	445151978.0	0.0	NaN
Film-Noir	3.0	104.000000	4.000000	100.0	102.00	104.0	106.00	108.0	3.0	7.966667	...	6.273068e+07	123353292.0	3.0	95.66666
Horror	11.0	102.090909	13.604812	71.0	98.00	103.0	109.00	122.0	11.0	7.909091	...	1.362817e+08	298791505.0	11.0	80.00000
Mystery	12.0	119.083333	14.475423	96.0	110.75	117.5	130.25	138.0	12.0	7.975000	...	1.310949e+08	474203697.0	8.0	79.12500
Thriller	1.0	108.000000	NaN	108.0	108.00	108.0	108.00	108.0	1.0	7.800000	...	1.755074e+07	17550741.0	1.0	81.00000
Western	4.0	148.250000	17.153717	132.0	134.25	148.0	162.00	165.0	4.0	8.350000	...	1.920000e+07	31800000.0	4.0	78.25000

14 rows × 40 columns

```
genres.sample(2,replace=True)
```



	Series_Title	Released_Year	Runtime	Genre	IMDB_Rating	Director	Star1	No_of_Votes	Gross	Metascore
944	Batoru rowaiaru	2000	114	Action	7.6	Kinji Fukasaku	Tatsuya Fujiwara	169091	195856489.0	81.0
625	Apocalypto	2006	139	Action	7.8	Mel Gibson	Gerardo Taracena	291018	50866635.0	68.0
991	Kelly's Heroes	1970	144	Adventure	7.6	Brian G. Hutton	Clint Eastwood	45338	1378435.0	50.0
300	Ben-Hur	1959	212	Adventure	8.1	William Wyler	Charlton Heston	219466	74700000.0	90.0
891	Incredibles 2	2018	118	Animation	7.6	Brad Bird	Craig T. Nelson	250057	608581744.0	80.0
389	The Iron Giant	1999	86	Animation	8.0	Brad Bird	Eli Marienthal	172083	23159305.0	85.0
536	All the President's Men	1976	138	Biography	7.9	Alan J. Pakula	Dustin Hoffman	103031	70600000.0	84.0
635	Walk the Line	2005	136	Biography	7.8	James Mangold	Joaquin Phoenix	234207	119519402.0	72.0
826	Barton Fink	1991	116	Comedy	7.7	Joel Coen	Ethan Coen	113240	6153939.0	69.0
732	Me and Earl and the Dying Girl	2015	105	Comedy	7.7	Alfonso Gomez-Rejon	Thomas Mann	123210	6743776.0	74.0
438	Touch of Evil	1958	95	Crime	8.0	Orson Welles	Charlton Heston	98431	2237659.0	99.0
222	Prisoners	2013	153	Crime	8.1	Denis Villeneuve	Hugh Jackman	601149	61002302.0	70.0
555	High Noon	1952	85	Drama	7.9	Fred Zinnemann	Gary Cooper	97222	9450000.0	89.0
314	Gone with the Wind	1939	238	Drama	8.1	Victor Fleming	George Cukor	290074	198676459.0	97.0
698	Willy Wonka & the Chocolate Factory	1971	100	Family	7.8	Mel Stuart	Gene Wilder	178731	4000000.0	67.0
698	Willy Wonka & the Chocolate Factory	1971	100	Family	7.8	Mel Stuart	Gene Wilder	178731	4000000.0	67.0
321	Das Cabinet des Dr. Caligari	1920	76	Fantasy	8.1	Robert Wiene	Werner Krauss	57428	337574718.0	NaN
321	Das Cabinet des Dr. Caligari	1920	76	Fantasy	8.1	Robert Wiene	Werner Krauss	57428	337574718.0	NaN
456	The Maltese Falcon	1941	100	Film-Noir	8.0	John Huston	Humphrey Bogart	148928	2108060.0	96.0
456	The Maltese Falcon	1941	100	Film-Noir	8.0	John Huston	Humphrey Bogart	148928	2108060.0	96.0
544	Night of the Living Dead	1968	96	Horror	7.9	George A. Romero	Duane Jones	116557	89029.0	89.0

```
genres.nunique()
```



	Series_Title	Released_Year	Runtime	IMDB_Rating	Director	Star1	No_of_Votes	Gross	Metascore
Genre									
Action	172	61	78	15	123	121	172	172	50
Adventure	72	49	58	10	59	59	72	72	33
Animation	82	35	41	11	51	77	82	82	29
Biography	88	44	56	13	76	72	88	88	40
Comedy	155	72	70	11	113	133	155	155	44
Crime	106	56	65	14	86	85	107	107	39
Drama	289	83	95	14	211	250	288	287	52
Family	2	2	2	1	2	2	2	2	2
Fantasy	2	2	2	2	2	2	2	2	0
Film-Noir	3	3	3	3	3	3	3	3	3
Horror	11	11	10	8	10	11	11	11	9
Mystery	12	11	10	8	10	11	12	12	7
Thriller	1	1	1	1	1	1	1	1	1
Western	4	4	4	4	2	2	4	4	4

```
# agg method
# passing dict
genres.agg(
    {
        'Runtime':'mean',
        'IMDB_Rating':'mean',
        'No_of_Votes':'sum',
        'Gross':'sum',
        'Metascore':'min'
    }
)
```



	Runtime	IMDB_Rating	No_of_Votes	Gross	Metascore
Genre					
Action	129.046512	7.949419	72282412	3.263226e+10	33.0
Adventure	134.111111	7.937500	22576163	9.496922e+09	41.0
Animation	99.585366	7.930488	21978630	1.463147e+10	61.0
Biography	136.022727	7.938636	24006844	8.276358e+09	48.0
Comedy	112.129032	7.901290	27620327	1.566387e+10	45.0
Crime	126.392523	8.016822	33533615	8.452632e+09	47.0
Drama	124.737024	7.957439	61367304	3.540997e+10	28.0
Family	107.500000	7.800000	551221	4.391106e+08	67.0
Fantasy	85.000000	8.000000	146222	7.827267e+08	NaN
Film-Noir	104.000000	7.966667	367215	1.259105e+08	94.0
Horror	102.090909	7.909091	3742556	1.034649e+09	46.0
Mystery	119.083333	7.975000	4203004	1.256417e+09	52.0
Thriller	108.000000	7.800000	27733	1.755074e+07	81.0
Western	148.250000	8.350000	1289665	5.822151e+07	69.0

```
# passing list
genres.agg(['min', 'max', 'mean', 'sum'])
```



Genre	Runtime				IMDB_Rating				No_of_Votes				Gross			
	min	max	mean	sum	min	max	mean	sum	min	max	mean	sum	min	max	mean	
Action	45	321	129.046512	22196	7.6	9.0	7.949419	1367.3	25312	2303232	420246.581395	72282412	3296.0	936662225.0	1.89	
Adventure	88	228	134.111111	9656	7.6	8.6	7.937500	571.5	29999	1512360	313557.819444	22576163	61001.0	874211619.0	1.31	
Animation	71	137	99.585366	8166	7.6	8.6	7.930488	650.3	25229	999790	268032.073171	21978630	128985.0	873839108.0	1.78	
Biography	93	209	136.022727	11970	7.6	8.9	7.938636	698.6	27254	1213505	272805.045455	24006844	21877.0	753585104.0	9.40	
Comedy	68	188	112.129032	17380	7.6	8.6	7.901290	1224.7	26337	939631	178195.658065	27620327	1305.0	886752933.0	1.01	
Crime	80	229	126.392523	13524	7.6	9.2	8.016822	857.8	27712	1826188	313398.271028	33533615	6013.0	790482117.0	7.89	
Drama	64	242	124.737024	36049	7.6	9.3	7.957439	2299.7	25088	2343110	212343.612457	61367304	3600.0	924558264.0	1.22	
Family	100	115	107.500000	215	7.8	7.8	7.800000	15.6	178731	372490	275610.500000	551221	4000000.0	435110554.0	2.19	
Fantasy	76	94	85.000000	170	7.9	8.1	8.000000	16.0	57428	88794	73111.000000	146222	337574718.0	445151978.0	3.91	
Film-Noir	100	108	104.000000	312	7.8	8.1	7.966667	23.9	59556	158731	122405.000000	367215	449191.0	123353292.0	4.19	
Horror	71	122	102.090909	1123	7.6	8.5	7.909091	87.0	27007	787806	340232.363636	3742556	89029.0	298791505.0	9.40	
Mystery	96	138	119.083333	1429	7.6	8.4	7.975000	95.7	33982	1129894	350250.333333	4203004	1035953.0	474203697.0	1.04	
Thriller	108	108	108.000000	108	7.8	7.8	7.800000	7.8	27733	27733	27733.000000	27733	17550741.0	17550741.0	1.75	
Western	132	165	148.250000	593	7.8	8.8	8.350000	33.4	65659	688390	322416.250000	1289665	5321508.0	31800000.0	1.45	

```
# Adding both the syntax
genres.agg(
    {
        'Runtime':['min','mean'],
        'IMDB_Rating':'mean',
        'No_of_Votes':['sum','max'],
        'Gross':'sum',
        'Metascore':'min'
    }
)
```



Genre	Runtime		IMDB_Rating	No_of_Votes		Gross	Metascore
	min	mean	mean	sum	max	sum	min
Action	45	129.046512	7.949419	72282412	2303232	3.263226e+10	33.0
Adventure	88	134.111111	7.937500	22576163	1512360	9.496922e+09	41.0
Animation	71	99.585366	7.930488	21978630	999790	1.463147e+10	61.0
Biography	93	136.022727	7.938636	24006844	1213505	8.276358e+09	48.0
Comedy	68	112.129032	7.901290	27620327	939631	1.566387e+10	45.0
Crime	80	126.392523	8.016822	33533615	1826188	8.452632e+09	47.0
Drama	64	124.737024	7.957439	61367304	2343110	3.540997e+10	28.0
Family	100	107.500000	7.800000	551221	372490	4.391106e+08	67.0
Fantasy	76	85.000000	8.000000	146222	88794	7.827267e+08	NaN
Film-Noir	100	104.000000	7.966667	367215	158731	1.259105e+08	94.0
Horror	71	102.090909	7.909091	3742556	787806	1.034649e+09	46.0
Mystery	96	119.083333	7.975000	4203004	1129894	1.256417e+09	52.0
Thriller	108	108.000000	7.800000	27733	27733	1.755074e+07	81.0
Western	132	148.250000	8.350000	1289665	688390	5.822151e+07	69.0

```
# looping on groups
df = pd.DataFrame(columns=movies.columns)
for group,data in genres:
```

```
df = df.append(data[data['IMDB_Rating'] == data['IMDB_Rating'].max()])
```

df



	Series_Title	Released_Year	Runtime	Genre	IMDB_Rating	Director	Star1	No_of_Votes	Gross	Metascore
2	The Dark Knight	2008	152	Action	9.0	Christopher Nolan	Christian Bale	2303232	534858444.0	84.0
21	Interstellar	2014	169	Adventure	8.6	Christopher Nolan	Matthew McConaughey	1512360	188020017.0	74.0
23	Sen to Chihiro no kamikakushi	2001	125	Animation	8.6	Hayao Miyazaki	Daveigh Chase	651376	10055859.0	96.0
7	Schindler's List	1993	195	Biography	8.9	Steven Spielberg	Liam Neeson	1213505	96898818.0	94.0
19	Gisaengchung	2019	132	Comedy	8.6	Bong Joon Ho	Kang-ho Song	552778	53367844.0	96.0
26	La vita è bella	1997	116	Comedy	8.6	Roberto Benigni	Roberto Benigni	623629	57598247.0	59.0
1	The Godfather	1972	175	Crime	9.2	Francis Ford Coppola	Marlon Brando	1620367	134966411.0	100.0
0	The Shawshank Redemption	1994	142	Drama	9.3	Frank Darabont	Tim Robbins	2343110	28341469.0	80.0
688	E.T. the Extra-Terrestrial	1982	115	Family	7.8	Steven Spielberg	Henry Thomas	372490	435110554.0	91.0
698	Willy Wonka & the Chocolate Factory	1971	100	Family	7.8	Mel Stuart	Gene Wilder	178731	4000000.0	67.0
321	Das Cabinet des Dr. Caligari	1920	76	Fantasy	8.1	Robert Wiene	Werner Krauss	57428	337574718.0	NaN
309	The Third Man	1949	104	Film-Noir	8.1	Carol Reed	Orson Welles	158731	449191.0	97.0

```
# split (apply) combine
# apply -> builtin function

genres.apply(min)
```




	Series_Title	Released_Year	Runtime	Genre	IMDB_Rating	Director	Star1	No_of_Votes	Gross	Metascore
Genre										
Action	300	1924	45	Action	7.6	Abhishek Chaubey	Aamir Khan	25312	3296.0	33.0
Adventure	2001: A Space Odyssey	1925	88	Adventure	7.6	Akira Kurosawa	Aamir Khan	29999	61001.0	41.0
Animation	Akira	1940	71	Animation	7.6	Adam Elliot	Adrian Molina	25229	128985.0	61.0
Biography	12 Years a Slave	1928	93	Biography	7.6	Adam McKay	Adrien Brody	27254	21877.0	48.0
Comedy	(500) Days of Summer	1921	68	Comedy	7.6	Alejandro G. Iñárritu	Aamir Khan	26337	1305.0	45.0
Crime	12 Angry Men	1931	80	Crime	7.6	Akira Kurosawa	Ajay Devgn	27712	6013.0	47.0
Drama	1917	1925	64	Drama	7.6	Aamir Khan	Abhay Deol	25088	3600.0	28.0
Family	E.T. the Extra-Terrestrial	1971	100	Family	7.8	Mel Stuart	Gene Wilder	178731	4000000.0	67.0
Fantasy	Das Cabinet des Dr. Caligari	1920	76	Fantasy	7.9	F.W. Murnau	Max Schreck	57428	337574718.0	NaN
Film-Noir	Shadow of a Doubt	1941	100	Film-Noir	7.8	Alfred Hitchcock	Humphrey Bogart	59556	449191.0	94.0

find number of movies starting with A for each group

```
def foo(group):
    return group['Series_Title'].str.startswith('A').sum()
```

genres.apply(foo)



Genre	
Action	10
Adventure	2
Animation	2
Biography	9
Comedy	14
Crime	4
Drama	21
Family	0
Fantasy	0
Film-Noir	0
Horror	1
Mystery	0
Thriller	0
Western	0

dtype: int64

find ranking of each movie in the group according to IMDB score

```
def rank_movie(group):
    group['genre_rank'] = group['IMDB_Rating'].rank(ascending=False)
    return group
```

genres.apply(rank_movie)

	Series_Title	Released_Year	Runtime	Genre	IMDB_Rating	Director	Star1	No_of_Votes	Gross	Metascore	genre_rank
0	The Shawshank Redemption	1994	142	Drama	9.3	Frank Darabont	Tim Robbins	2343110	28341469.0	80.0	1.0
1	The Godfather	1972	175	Crime	9.2	Francis Ford Coppola	Marlon Brando	1620367	134966411.0	100.0	1.0
2	The Dark Knight	2008	152	Action	9.0	Christopher Nolan	Christian Bale	2303232	534858444.0	84.0	1.0
3	The Godfather: Part II	1974	202	Crime	9.0	Francis Ford Coppola	Al Pacino	1129952	57300000.0	90.0	2.5
4	12 Angry Men	1957	96	Crime	9.0	Sidney Lumet	Henry Fonda	689845	4360000.0	96.0	2.5
...	...	...	...	...	...	...	...	...	...	...	...
995	Breakfast at Tiffany's	1961	115	Comedy	7.6	Blake Edwards	Audrey Hepburn	166544	679874270.0	76.0	147.0
996	Giant	1956	201	Drama	7.6	George Stevens	Elizabeth Taylor	34075	195217415.0	84.0	272.5

```
# find normalized IMDB rating group wise
```

```
def normal(group):
    group['norm_rating'] = (group['IMDB_Rating'] - group['IMDB_Rating'].min())/(group['IMDB_Rating'].max() - group['IMDB_Rating'].min())
    return group
```

```
genres.apply(normal)
```

	Series_Title	Released_Year	Runtime	Genre	IMDB_Rating	Director	Star1	No_of_Votes	Gross	Metascore	norm_rating
0	The Shawshank Redemption	1994	142	Drama	9.3	Frank Darabont	Tim Robbins	2343110	28341469.0	80.0	1.000
1	The Godfather	1972	175	Crime	9.2	Francis Ford Coppola	Marlon Brando	1620367	134966411.0	100.0	1.000
2	The Dark Knight	2008	152	Action	9.0	Christopher Nolan	Christian Bale	2303232	534858444.0	84.0	1.000
3	The Godfather: Part II	1974	202	Crime	9.0	Francis Ford Coppola	Al Pacino	1129952	57300000.0	90.0	0.875
4	12 Angry Men	1957	96	Crime	9.0	Sidney Lumet	Henry Fonda	689845	4360000.0	96.0	0.875
...	...	...	...	...	...	...	...	...	...	...	...
995	Breakfast at Tiffany's	1961	115	Comedy	7.6	Blake Edwards	Audrey Hepburn	166544	679874270.0	76.0	0.000

```
# groupby on multiple cols
duo = movies.groupby(['Director', 'Star1'])
duo
# size
duo.size()
# get_group
duo.get_group(('Aamir Khan', 'Amole Gupte'))
```

↗

	Series_Title	Released_Year	Runtime	Genre	IMDB_Rating	Director	Star1	No_of_Votes	Gross	Metascore
65	Taare Zameen Par	2007	165	Drama	8.4	Aamir Khan	Amole Gupte	168895	1223869.0	NaN

```
# find the most earning actor->director combo
duo['Gross'].sum().sort_values(ascending=False).head(1)
```

↗

Director	Star1	
Akira Kurosawa	Toshirô Mifune	2.999877e+09
Name: Gross, dtype: float64		

```
# find the best(in-terms of metascore(avg)) actor->genre combo
movies.groupby(['Star1', 'Genre'])['Metascore'].mean().reset_index().sort_values('Metascore', ascending=False).head(1)
```

↗

	Star1	Genre	Metascore
230	Ellar Coltrane	Drama	100.0

```
# agg on multiple groupby
duo.agg(['min', 'max', 'mean'])
```

↗

		Runtime			IMDB_Rating			No_of_Votes			Gross			Metascore		
		min	max	mean	min	max	mean	min	max	mean	min	max	mean	min	max	mean
Director	Star1															
Aamir Khan	Amole Gupte	165	165	165.0	8.4	8.4	8.4	168895	168895	168895.0	1223869.0	1223869.0	1223869.0	NaN	NaN	NaN
Aaron Sorkin	Eddie Redmayne	129	129	129.0	7.8	7.8	7.8	89896	89896	89896.0	853090410.0	853090410.0	853090410.0	77.0	77.0	77.0
Abdellatif Kechiche	Léa Seydoux	180	180	180.0	7.7	7.7	7.7	138741	138741	138741.0	2199675.0	2199675.0	2199675.0	89.0	89.0	89.0
Abhishek Chaubey	Shahid Kapoor	148	148	148.0	7.8	7.8	7.8	27175	27175	27175.0	218428303.0	218428303.0	218428303.0	NaN	NaN	NaN
Abhishek Kapoor	Amit Sadh	130	130	130.0	7.7	7.7	7.7	32628	32628	32628.0	1122527.0	1122527.0	1122527.0	40.0	40.0	40.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Zaza Urushadze	Lembit Ulfsak	87	87	87.0	8.2	8.2	8.2	40382	40382	40382.0	144501.0	144501.0	144501.0	73.0	73.0	73.0
Zoya Akhtar	Hrithik Roshan	155	155	155.0	8.1	8.1	8.1	67927	67927	67927.0	3108485.0	3108485.0	3108485.0	NaN	NaN	NaN
		151	151	151.0	8.0	8.0	8.0	81888	81888	81888.0	5500581.0	5500581.0	5500581.0	85.0	85.0	85.0

↕ Exerciscie

```
ipl = pd.read_csv('/content/deliveries.csv')
ipl.head()
```

	match_id	inning	batting_team	bowling_team	over	ball	batsman	non_striker	bowler	is_super_over	...	bye_runs	legbye_runs	nob
0	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	1	DA Warner	S Dhawan	TS Mills	0	...	0	0	
1	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	2	DA Warner	S Dhawan	TS Mills	0	...	0	0	
2	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	3	DA Warner	S Dhawan	TS Mills	0	...	0	0	
3	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	4	DA Warner	S Dhawan	TS Mills	0	...	0	0	
4	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	5	DA Warner	S Dhawan	TS Mills	0	...	0	0	

```
ipl.shape
```

```
(179078, 21)
```

```
# find the top 10 batsman in terms of runs
```

```
ipl.groupby('batsman')['batsman_runs'].sum().sort_values(ascending=False).head(10)
```

```
batsman
V Kohli      5434
SK Raina     5415
RG Sharma    4914
DA Warner    4741
S Dhawan     4632
CH Gayle     4560
MS Dhoni     4477
RV Uthappa   4446
AB de Villiers 4428
G Gambhir    4223
Name: batsman_runs, dtype: int64
```

```
# find the batsman with max no of sixes
```

```
six = ipl[ipl['batsman_runs'] == 6]
```

```
six.groupby('batsman')['batsman'].count().sort_values(ascending=False).head(1).index[0]
```

```
'CH Gayle'
```